

Cryptocurrency & Blockchain Basics





The Missing Cricket Bat Record

You and 4 friends rent a cricket bat for ₹100.
Who holds the money record? What if the
record keeper loses the note – or secretly
changes ₹100 to ₹50?



What is a Ledger? The Khata Book

A ledger is a notebook that records who gave money to whom. Just like the Khata book at your local Kirana store keeps track of every transaction – a blockchain ledger does the same, but digitally!



Kirana store's Khata book tracks credit & payments



Class attendance register records who is present



Bank passbook shows all your money transactions





The Problem with One Record Keeper

Imagine only the teacher has the grade book. What happens if something goes wrong? One person in charge means one big risk!



Only the teacher holds the grade book – centralized record!



The record can get lost or damaged easily



Grades could be secretly changed – no one would know!

The Magic Shared Register

Everyone has an identical copy of the record book. If one changes, others catch it! That's decentralization – no single person controls the record.



The 5-Friend Network Activity

Ananya sends ₹50 to Kabir. Watch what happens when everyone in the network records and verifies the transaction together!



Ananya sends ₹50 to Kabir – everyone writes it down!



All 5 friends check and verify the entry



Everyone agrees? ☒ Transaction confirmed!



How Blockchain Works - Building Blocks

Blockchain is made of connected blocks
– like pages in a magical register that
no one can secretly change!



DATA BLOCK = A page of transactions



DIGITAL FINGERPRINT (Hash) = A magical
stamp on each page



THE LINK = Each page stamps the
previous page's fingerprint



The Chain Connection

Each block holds the fingerprint of the one before it. Like envelopes tied with red Mauli thread – tamper one and the whole chain breaks!



Page 2 includes fingerprint of Page 1



Page 3 includes fingerprint of Page 2



Creates an unbreakable chain!





The Hacker Attack Simulation

Vivaan tries to change ₹50 to ₹500 in his notebook – but the blockchain catches him! Everyone checks and rejects the fake entry.



Vivaan changes ₹50 → ₹500 secretly!



His fingerprint changes & Page 2's link breaks!



4 friends check and reject the fake entry!

Cryptocurrency vs Blockchain

Think of Blockchain as a secure highway (like NH-44) and Cryptocurrency as the cars that drive on it. Blockchain can record much more than just money!



Blockchain = Secure highway (NH-44)



Cryptocurrency = Cars on the highway



Records marksheets, land records & mango supply chains!





Real-World Indian Examples

Blockchain is already part of everyday Indian life! From payments to mangoes, let's see where it shows up around us.



UPI payments – fast, verified digital transactions



IPL Auction Board – a transparent public ledger



Devgad Alphonso Mango tracked farm to supermarket



Quick Recap Quiz!

Let's see what you remember! Think carefully and pick the best answer for each question below.



What do we call a notebook that records who gave money to whom? A) Dictionary B) Khata/Ledger C) Atlas



If EVERYONE has a copy of the record book, what type of system is it? A) Centralized B) Secret C) Decentralized



What connects each block in a Blockchain? A) Red thread B) Digital fingerprint (Hash) C) A password



**Thank You!
So Much...**

